THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(a)** 2	$2250 = 2 \times 3^2 \times 5^3$
2	**(a)** 1	Each point where the graph meets the x-axis is a zero
3	**(c)** infinitely many solutions (coincident lines)	Compare a_1/a_2, b_1/b_2, c_1/c_2
4	**(b)** 7	Sum $= -b/a$
5	**(b)** -6	$a_n = a + (n-1)d$
6	**(c)** 6	BPT
7	**(b)** 5	Distance formula
8	**(b)** $\frac{4}{5}$	Hypotenuse is 5
9	**(b)** $15\sqrt{3}$ m	$h = 15\tan 60^\circ = 15 \times \sqrt{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 15\sqrt{3}$ m
10	**(a)** 50°	$\angle AOB = 100^\circ$ and OP bisects it
11	**(d)** ₹ 880	Cost $= 88 \times 10$; Cost = ₹ 880
12	**(d)** 858 cm²	TSA = CSA of cone + CSA of hemisphere $= \pi rl + 2\pi r^2$; TSA $= \frac{22}{7} \times 7 \times 25 + 2 \times \frac{22}{7} \times 7^2 = 858$ cm²
13	**(a)** 44.2	$\sum f_i = 50$ and $\sum f_i x_i = 2210$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{2210}{50} = 44.2$
14	**(b)** $\frac{5}{36}$	Favourable outcomes: (2,6), (3,5), (4,4), (5,3), (6,2); number $= 5$; $P = \frac{5}{36}$
15	**(d)** 22338	LCM $= \dfrac{306 \times 657}{9}$
16	**(a)** 5	Sum of zeroes $= 7$, so other $= 7 - 2 = 5$
17	**(a)** 5	Solve by elimination
18	**(c)** 9	$D = 0$: $4m^2 - 4k = 0$
19	**(a)** 84	Reverse AP has difference $-d$
20	**(a)** 9	Sides in ratio $\sqrt{\text{areas}}$
21	**(a)** $(2, 3)$	Section formula
22	**(a)** $\frac{12}{5}$	Find the perpendicular
23	**(d)** $3\sqrt{3}$ m	Height $= 6\sin 60^\circ = 6 \times \frac{\sqrt{3}}{2}$; Height reached on the wall $= 3\sqrt{3}$ m
24	**(a)** 30 cm	$PT^2 = OP^2 - OT^2 = 34^2 - 16^2 = 900$; $PT = 30$ cm
25	**(c)** 50	Distance 66 m $= 6600$ cm; revolutions $= \frac{6600}{132}$; Revolutions $= 50$
26	**(a)** 374 cm²	$2\pi rh + 2(2\pi r^2) = 2 \times \frac{22}{7} \times 3.5 \times 10 + 4 \times \frac{22}{7} \times 3.5^2$; TSA $= 374$ cm²
27	**(c)** 78.75	Median class 70–80: $l = 70$, $cf = 9$, $f = 4$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 70 + \frac{12.5-9}{4} \times 10 = 78.75$
28	**(c)** $\frac{3}{8}$	Favourable outcomes: HHT, HTH, THH; number $= 3$; $P = \frac{3}{8}$
29	**(d)** always irrational	If it were rational, dividing by the rational would make the irrational rational
30	**(b)** $x^2 - 7x + 12$	$x^2 - (\text{sum})x + \text{product}$
31	**(c)** 2	Solve by elimination
32	**(c)** 14	Other root 7; product $= k$
33	**(a)** 3	$a_{n+1} - a_n = 3$
34	**(b)** 18	$d^2 = s^2 + s^2$
35	**(d)** 12	$\frac{1}{2} \times$ base $\times$ height
36	**(a)** 1	Substitute the standard values
37	**(c)** 12 m	Add the height of the eyes: $2 + 10$; Height of building $= 12$ m
38	**(b)** 5 cm	$9 + 7 = AD + 11$; $AD = 5$ cm

39	**(d)** 28 cm	$R = r_1 + r_2 = 19 + 9$; $R = 28$ cm
40	**(a)** 1330 cm²	$6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$ with $r = 7$ cm; TSA $= 1176 + 154 = 1330$ cm^2
41	**(b)** 72.5	$l = 70$, $f_1 = 13$, $f_0 = 10$, $f_2 = 4$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 70 + \frac{3}{12} \times 10 = 72.5$
42	**(c)** $\frac{4}{11}$	Favourable letters: A, E, A, I; number $= 4$; $P = \frac{4}{11}$
43	**(b)** 5	$3125 = 5^5$; the number of places is the larger of the powers of 2 and 5
44	**(c)** 9	$\alpha + \beta = -b/a$
45	**(c)** 24	$x = 6$, $y = 4$
46	**(a)** 3 and 5	Factorise
47	**(a)** 3	$a_1 = S_1$
48	**(c)** 25	Pythagoras
49	**(b)** 8	$x = 2m - a$
50	**(c)** 1	$\tan(90^\circ - \theta) = \cot\theta$

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** $10\sqrt{3}$ m	$h(\cot 30^\circ - \cot 60^\circ) = 20$ gives $h = 20 \div (\frac{2\sqrt{3}}{3})$; Height $= 10\sqrt{3}$ m (distances 30 m and 10 m)
2	**(b)** 30°	$\angle OAP = 90^\circ$ (radius $\perp$ tangent); $\angle OAB = 90^\circ - 60^\circ = 30^\circ$
3	**(c)** 10 cm	$R^2 = 6^2 + 8^2 = 100$; $R = 10$ cm
4	**(a)** 1760 cm²	TSA = CSA cylinder + one flat base + CSA of cone; $= 1056 + 154 + 550 = 1760$ cm^2
5	**(d)** 30–40	It belongs to the class 30–40; Modal class = 30–40
6	**(b)** $\frac{1}{2}$	Favourable outcomes = number of red balls; $P = \frac{6}{12} = \frac{1}{2}$
7	**(c)** $\sqrt{5} + 3$	Rational + irrational is irrational
8	**(a)** 14	$\alpha\beta = c/a$
9	**(c)** 4	$\frac{1}{2} = \frac{2}{k} \neq \frac{3}{7}$
10	**(a)** 25	$D = b^2 - 4ac$
11	**(d)** 36	$S_n = n^2$
12	**(b)** right-angled	Check $9^2 + 12^2$ against 15^2
13	**(a)** 5	Distance from y-axis $= \lvert x \rvert$
14	**(b)** 1	$\cos(90^\circ - \theta) = \sin\theta$
15	**(d)** 12 m	Distance from the foot $= 4\cot 30^\circ = 4\sqrt{3}$ m; Original height of the tree $= 12$ m (that is $4 + 8$)
16	**(c)** 41 cm	$OP^2 = r^2 + PT^2 = 9^2 + 40^2 = 1681$; $OP = 41$ cm
17	**(a)** 115.5 cm²	Area $= \frac{30}{360} \times \frac{22}{7} \times 21^2$; Area $= 115.5$ cm^2
18	**(d)** 374 cm²	$2\pi rh + 2(2\pi r^2)$ with $r = 3.5$, $h = 10$; TSA $= 220 + 154 = 374$ cm^2
19	**(a)** 16	$\frac{1650+35p}{34+p} = 44.2$ gives $(35 - 44.2)p = -147.2$; $p = 16$
20	**(d)** $\frac{3}{10}$	Favourable numbers: 3, 6, 9, 12, 15, 18; number = 6; $P = \frac{6}{20} = \frac{3}{10}$
21	**(d)** no natural number n	$3 = 3$ lacks the prime factor 5
22	**(c)** 2 and 3	$(x - 2)(x - 3) = 0$ gives $x = 2$ or $x = 3$
23	**(c)** 2	Solve, then add
24	**(b)** two distinct real roots	$D = b^2 - 4ac = 12$
25	**(c)** 5	$a + 9d$
26	**(b)** 5	$\frac{AE}{AC} = \frac{AD}{AB}$
27	**(c)** 3	Equate the squared distances
28	**(a)** 4	$k^2(\sin^2 + \cos^2)$
29	**(d)** $7 + 7\sqrt{3}$ m	Part of the tower above the top of the building $= d\tan 60^\circ = 7\sqrt{3}$ m; Tower $= 7 + 7\sqrt{3}$ m
30	**(a)** 2 cm	$AF = AB - r = 6 - r$, $CD = BC - r = 8 - r$ and $AF + CD = AC$ (tangent lengths); $(6 - r) + (8 - r) = 10$ gives $r = 2$ cm
31	**(d)** 38.5 m²	Area $= \frac{90}{360} \times \frac{22}{7} \times 7^2$; Area $= 38.5$ m^2
32	**(a)** 32	Volume of a sphere $= \frac{4}{3}\pi(3)^3$; $n = \frac{3r^2h}{4\rho^3}$; $n = 32$
33	**(b)** 17.5	$\frac{10+25}{2}$
34	**(b)** $\frac{8}{9}$	Favourable outcomes = number of good pens; $P = \frac{120}{135} = \frac{8}{9}$
35	**(a)** 5	$480 = 2^5 \times 3 \times 5$
36	**(b)** 7	Sum $= -b/a$
37	**(a)** no solution (parallel lines)	Compare a_1/a_2, b_1/b_2, c_1/c_2

38	**(c)** 9	Sum $= -b/a$
39	**(d)** 78	$a_n = a + (n-1)d$
40	**(b)** 12	BPT
41	**(c)** 13	Distance formula
42	**(a)** $\frac{3}{5}$	Reciprocal
43	**(d)** 45°	$\tan\theta = 1$ which is $\tan 45^\circ$; $\theta = 45^\circ$
44	**(c)** 17 cm	$3x + 5 = 2x + 9$ gives $x = 4$; $QA = 3(4) + 5 = 17$ cm
45	**(b)** 60°	$\theta = \frac{231 \times 360}{\frac{22}{7} \times 21^2}$; $\theta = 60^\circ$
46	**(c)** 6 cm	$\frac{4}{3}\pi R^3 = \frac{1}{3}\pi(6)^2(24)$ gives $R^3 = 216$; $R = 6$ cm
47	**(a)** 23.8	$\sum f_i = 50$ and $\sum f_i x_i = 1190$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{1190}{50} = 23.8$
48	**(a)** $\frac{1}{6}$	Favourable outcomes: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1); number $= 6$; $P = \frac{6}{36} = \frac{1}{6}$
49	**(a)** 2	$3600 = 2^4 \times 3^2 \times 5^2$
50	**(c)** 2	Each point where the graph meets the x-axis is a zero

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(a)** 4	Solve by elimination
2	**(b)** 16	$D = 0$: $4m^2 - 4k = 0$
3	**(a)** 59	Reverse AP has difference $-d$
4	**(d)** 12	Sides in ratio $\sqrt{\text{areas}}$
5	**(a)** $(5,\ 6)$	Section formula
6	**(b)** $\frac{12}{13}$	Hypotenuse is 13
7	**(c)** $7\sqrt{3}$ m	$h = 21\tan 30^\circ = 21 \times \frac{\sqrt{3}}{3}$ (with $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 60^\circ = \sqrt{3}$); Height $= 7\sqrt{3}$ m
8	**(d)** $\sqrt{119}$ cm	$PQ^2 = 144 - 25 = 119$
9	**(b)** ₹ 660	Cost $= 132 \times 5$; Cost = ₹ 660
10	**(c)** 5082 cm²	TSA = CSA of cone + CSA of hemisphere $= \pi rl + 2\pi r^2$; TSA $= \frac{22}{7} \times 21 \times 35 + 2 \times \frac{22}{7} \times 21^2 = 5082$ cm^2
11	**(d)** 65	Median class 60–70: $l = 60$, $cf = 19$, $f = 12$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 60 + \frac{25-19}{12} \times 10 = 65$
12	**(a)** $\frac{1}{2}$	Favourable outcomes: HHH, HHT, HTH, THH; number $= 4$; $P = \frac{4}{8} = \frac{1}{2}$
13	**(b)** 23460	LCM $= \dfrac{510 \times 92}{2}$
14	**(d)** 6	Sum of zeroes $= 9$, so other $= 9 - 3 = 6$
15	**(c)** 1	Solve by elimination
16	**(d)** 24	Other root 8; product $= k$
17	**(c)** 3	$a_{n+1} - a_n = 3$
18	**(b)** 50	$d^2 = s^2 + s^2$
19	**(c)** 20	$\frac{1}{2}\times$ base $\times$ height
20	**(b)** $\frac{15}{8}$	Find the perpendicular
21	**(b)** $8\sqrt{3}$ m	Height $= 16\sin 60^\circ = 16 \times \frac{\sqrt{3}}{2}$; Height reached on the wall $= 8\sqrt{3}$ m
22	**(a)** 28 cm	$PT^2 = OP^2 - OT^2 = 35^2 - 21^2 = 784$; $PT = 28$ cm
23	**(d)** 50	Distance 110 m = 11000 cm; revolutions $= \frac{11000}{220}$; Revolutions $= 50$
24	**(c)** 1232 cm²	$2\pi rh + 2(2\pi r^2) = 2 \times \frac{22}{7} \times 7 \times 14 + 4 \times \frac{22}{7} \times 7^2$; TSA $= 1232$ cm^2
25	**(d)** 85.5	$l = 80$, $f_1 = 16$, $f_0 = 5$, $f_2 = 7$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 80 + \frac{11}{20} \times 10 = 85.5$
26	**(c)** $\frac{2}{11}$	Favourable letters: B, B; number $= 2$; $P = \frac{2}{11}$
27	**(a)** always irrational	If it were rational, dividing by the rational would make the irrational rational
28	**(d)** $x^2 - 3x - 10$	$x^2 - (\text{sum})x + \text{product}$
29	**(d)** 45	$x = 9$, $y = 5$
30	**(d)** -4 and 6	Factorise
31	**(b)** 4	$a_1 = S_1$
32	**(d)** 40	Pythagoras
33	**(b)** 10	$x = 2m - a$
34	**(c)** $\frac{8}{3}$	Substitute the standard values
35	**(b)** 18 m	Add the height of the eyes: $3 + 15$; Height of building $= 18$ m
36	**(b)** 10 cm	$12 + 8 = AD + 10$; $AD = 10$ cm
37	**(a)** 14 cm	$R = r_1 + r_2 = 8 + 6$; $R = 14$ cm
38	**(a)** 5320 cm²	$6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$ with $r = 14$ cm; TSA $= 4704 + 616 = 5320$ cm^2

39	**(b)** 30–40	It belongs to the class 30–40; Modal class $= 30$–40
40	**(c)** $\frac{2}{3}$	Favourable outcomes = number of red balls; $P = \frac{10}{15} = \frac{2}{3}$
41	**(b)** 3	$8 = 2^3$; the number of places is the larger of the powers of 2 and 5
42	**(b)** 5	$\alpha + \beta = -b/a$
43	**(b)** 9	$\frac{1}{3} = \frac{3}{k} \neq \frac{4}{13}$
44	**(b)** -56	$D = b^2 - 4ac$
45	**(c)** 64	$S_n = n^2$
46	**(b)** not right-angled	Check $7^2 + 8^2$ against 10^2
47	**(d)** 7	Distance from y-axis $= \lvert x \rvert$
48	**(b)** 1	$\tan(90^\circ - \theta) = \cot\theta$
49	**(a)** $15\sqrt{3}$ m	$h(\cot 30^\circ - \cot 60^\circ) = 30$ gives $h = 30 \div (\frac{2\sqrt{3}}{3})$; Height $= 15\sqrt{3}$ m (distances 45 m and 15 m)
50	**(b)** 35°	$\angle OAP = 90^\circ$ (radius $\perp$ tangent); $\angle OAB = 90^\circ - 55^\circ = 35^\circ$

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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